

1)

Current: Anya Neighbors: Frog, Dog, Penguin, Whale

Node	Distance	Prev
Anya	0	
Frog	0.1	Anya
Dog	4	Anya
Penguin	2	Anya
Whale	1	Anya

Visited: Anya Priority Queue: Frog, Dog, Penguin, Whale
(All other nodes have infinite distance)

Current: Frog Neighbors: None

Visited: Anya, Frog PQ: Dog, Penguin, Whale

Current: Dog Neighbors: Bunny

Node	Distance	Prev
Dog	4	Anya
Bunny	7	Dog

Visited: Anya, Frog, Dog, PQ: Penguin, Whale, Bunny

Current: Penguin Neighbor: Dog

Node	Distance	Prev
Penguin	2	Anya
Dog	4	Penguin (unchange)

Visited: Anya, Frog, Dog, Penguin, PQ: Whale, Bunny

Current: Whale Neighbors: Elephant, Wolf, Rhino

Node	Distance	Prev
Whale	1	Anya
Elephant	4	Whale
Wolf	6	Whale
Rhino	2	Whale

Visited: Anya, Frog, Dog, Penguin, Whale, PQ: Bunny, Elephant, Wolf, Rhino

1)

Current: Bunny Neighbors: Bond

Node	Distance	Prev
Bunny	7	Dog
Bond	12	Bunny

Visited: Anya, Frog, Dog, Penguin, Whale, Bunny

PQ: Elephant, Wolf, Rhino, Bond

Current: Elephant Neighbors: Penguin

Node	Distance	Prev
Elephant	4	Whale
Penguin	7	Elephant

V: Anya, Frog, Dog, Penguin, Whale, Bunny, Elephant

PQ: Wolf, Rhino, Bond

Current: Wolf Neighbors: None

V: Anya, Frog, Dog, Penguin, Whale, Bunny, Elephant, Wolf

PQ: Rhino, Bond

Current: Rhino Neighbors: Panda, Elephant

Node	Distance	Prev
Rhino	2	Whale
Panda	14	Rhino
Elephant	2.5	Rhino (update)

V: Anya, Frog, Dog, Pen, Whale, Bunny, Ele, Wolf, Rhin

PQ Bond, Panda

Current: Bond Neighbor: Panda, dog

Node	Distance	Prev
Bond	12	Bunny
Dog	13	Bond
Panda	13.5	Bond (update)

V: Anya, Frog, Dog, Pe, Whale, Bunny, Ele, Wolf, Rhino, Bond

PQ Panda

1)

▷ Current: Panda Neighbors: Bond, Sloth

Node	Distance	Prev
Panda	13.5	Bond
Bond	15.2	Panda
Sloth	14.	Panda

▷ V: Anty, Frog, Dog, Pig, Whale, Ele, Wolf, Rhino, Bunny, Bird, Panda
▷ PQ: Sloth

▷ Current: Sloth Neighbors: Bond, Elephant

Node	Distance	Prev
Sloth	14	Panda
Bond	18	Sloth
Elephant	15	Sloth

▷ V: All
▷ PQ: None!

1)

Final table

Node	Distance	Prev
Anya	0	
Frog	0.1	Anya
Dog	4	Anya
Penguin	2	Anya
Whale	1	Anya
Elephant	2.5	Rhino
Wolf	6	Whale
Rhino	2	Whale
Bunny	7	Dog
Bond	12	Bunny
Panda	13.5	Bond
Sloth	14	Panda

Paths:

Anya

Anya → Frog

Anya → Dog

Anya → Penguin

Anya → Whale

Anya → Whale → Rhino

Anya → Whale → Rhino → Elephant

Anya → Whale → Wolf

Anya → Dog → Bunny

Anya → Dog → Bunny → Bond

Anya → Dog → Bunny → Bond → Panda

Anya → Dog → Bunny → Bond → Panda → Sloth

2)

A star

0)

	0	1	2	3	4	5	6
0	Bunny (0,0)			Bond (3,0)		Panda (5,0)	
1				Sloth (3,1)			Rhino (6,1)
2		Dog (1,2)	Penguin (2,2)		Elephant (4,2)	Wolf (5,2)	
3	Frog (0,3)		Anya (2,3)				Whale (6,3)

2)

Current: Anya Neighbors: Frog, Dog, Penguin, Whale

Node	g	h	f	Prev
Anya	0	0	0	
Frog	0.1	2	2.1	Anya
Dog	4	2	6	Anya
Penguin	2	1	3	Anya
Whale	1	4	5	Anya

Visited: Anya Priority Queue: Frog, Dog, Penguin, Whale

Current: Frog Neighbors: None
Visited: Anya, Frog PQ: Dog, Penguin, Whale

Current: Dog Neighbors: Bunny

Node	g	h	f	Prev
Dog	4	2	6	Anya
Bunny	3	3	6	Dog

Visited: Anya, Frog, Dog PQ: Penguin, Whale, Bunny

Current: Penguin Neighbors: Dog

Node	g	h	f	Prev
Penguin	2	1	3	Anya
Dog	2	1	3	Penguin

Visited: Anya, Frog, Dog, Penguin PQ: Whale, Bunny.

Current: Whale Neighbors: Elephant, Wolf, Rhino

Node	g	h	f	Prev
Whale	1	4	5	Anya
Elephant	3	3	6	Whale
Wolf	5	2	7	Whale
Rhino	1	2	3	Whale

Visited: Anya, Frog, Dog, Penguin, Whale PQ: Bunny, Elephant, Rhino

2)

▷ Current: Bunny Neighbors: Bond, Frog

Node	g	h	f	Prev
Bunny	3	3	6	Dog
Bond	5	3	8	Bunny
Frog	2	2	4	Bunny

▷ Visited: Anya, Frog, Dog, Penguin, Whale, Bunny

▷ PQ: Elephant, Wolf, Rhino, Bond

▷ Current: Elephant Neighbors: Penguin

Node	g	h	f	Prev
Elephant	3	3	6	Whale
Penguin	3	2	5	Elephant

▷ Visited: Anya, Frog, Dog, Penguin, Whale, Bunny, Elephant

▷ PQ: Wolf, Rhino, Bond

▷ Current: Wolf Neighbors: None

▷ Visited: Anya, Frog, Dog, Penguin, Whale, Bunny, Elephant, Wolf

▷ PQ: Rhino, Bond

▷ Current: Rhino Neighbors: Panda, Elephant

Node	g	h	f	Prev
Rhino	1	2	3	Whale
Panda	12	2	14	Rhino
Elephant	0.5	3	3.5	Rhino

▷ Visited: Anya, Frog, Dog, Penguin, Whale, Bunny, Elephant, Wolf, Rhino

▷ PQ: Bond, Panda

2)

Current: Bond Neighbors: Panda, Dog

Node	g	h	f	Prev
Bond	5	3	8	Bunny
Panda	1.5	2	3.5	Bond
Dog	1	4	5	Bond

Visited: Anga, Frog, Dog, Penguin, Whale, Bunny, Elephant, Wolf
 Rhino, Bond
 PQ: Panda

Current: Panda Neighbors: Sloth, Bond

Node	g	h	f	Prev
Panda	1.5	2	3.5	Bond
Sloth	0.5	3	3.5	Panda
Bond	1.7	2	3.7	Panda

Visited: Anga, Frog, Dog, Penguin, Whale, Bunny, Elephant
 Wolf, Rhino, Bond, Panda
 PQ: Sloth

Current: Sloth Neighbors: Bond, Elephant, Wolf

Node	g	h	f	Prev
Sloth	0.5	3	3.5	Panda
Bond	4	1	5	Sloth
Elephant	1	2	3	Sloth
Wolf	1	1	2	Sloth

Final table

Node	g	h	f	Prev
Anya	0	0	0	
Frog	0.1	2	2.1	Anya
Dog	4	2	6	Anya
Penguin	2	1	3	Anya
Whale	1	4	5	Anya
Bunny	3	3	6	Dog
Elephant	3	3	6	Whale
Wolf	5	2	7	Whale
Rhino	1	2	3	Whale
Bond	5	3	8	Bunny
Panda	1.5	2	3.5	Bond
Sloth	0.5	3	3.5	Panda

Final Path

Anya
 Anya → Frog
 Anya → Dog
 Anya → Penguin
 Anya → Whale
 Anya → Dog → Bunny
 Anya → Whale → elephant
 Anya → Whale → Wolf
 Anya → Whale → Rhino
 Anya → Dog → Bunny → Bond
 Anya → Whale → Rhino → Panda
 Anya → Whale → Rhino → Panda → Sloth.

3) Dijkstra and A* both can be quicker than each other, for example Dijkstra finds elephant faster than A* finds elephant. But A* finds bunny faster than Dijkstra.

4)

Time complexity is $O(n^2)$, where n is the number of nodes in the adjacency matrix, because the BFS outer loop runs once for each node n times, and for each node we loop through all other nodes to check for edges.

The space complexity is $O(n)$, where n is the number of nodes in the graph, because we create a boolean array of size n and a queue that holds at most n nodes at any given time.

5)

6)

Time complexity is $O(n)$, where n is the number of vertices in the input string/string array created, because the draw method has a single loop that iterates through all n vertices to build the matrix. Inside the loop all the calculations are in constant time operations.

The space complexity is $O(n^2)$, where n is the number of vertices in the input string, because the adjacency matrix we create is of size n times n . Everything else created is $O(n)$ or less which means we can ignore the lower order terms